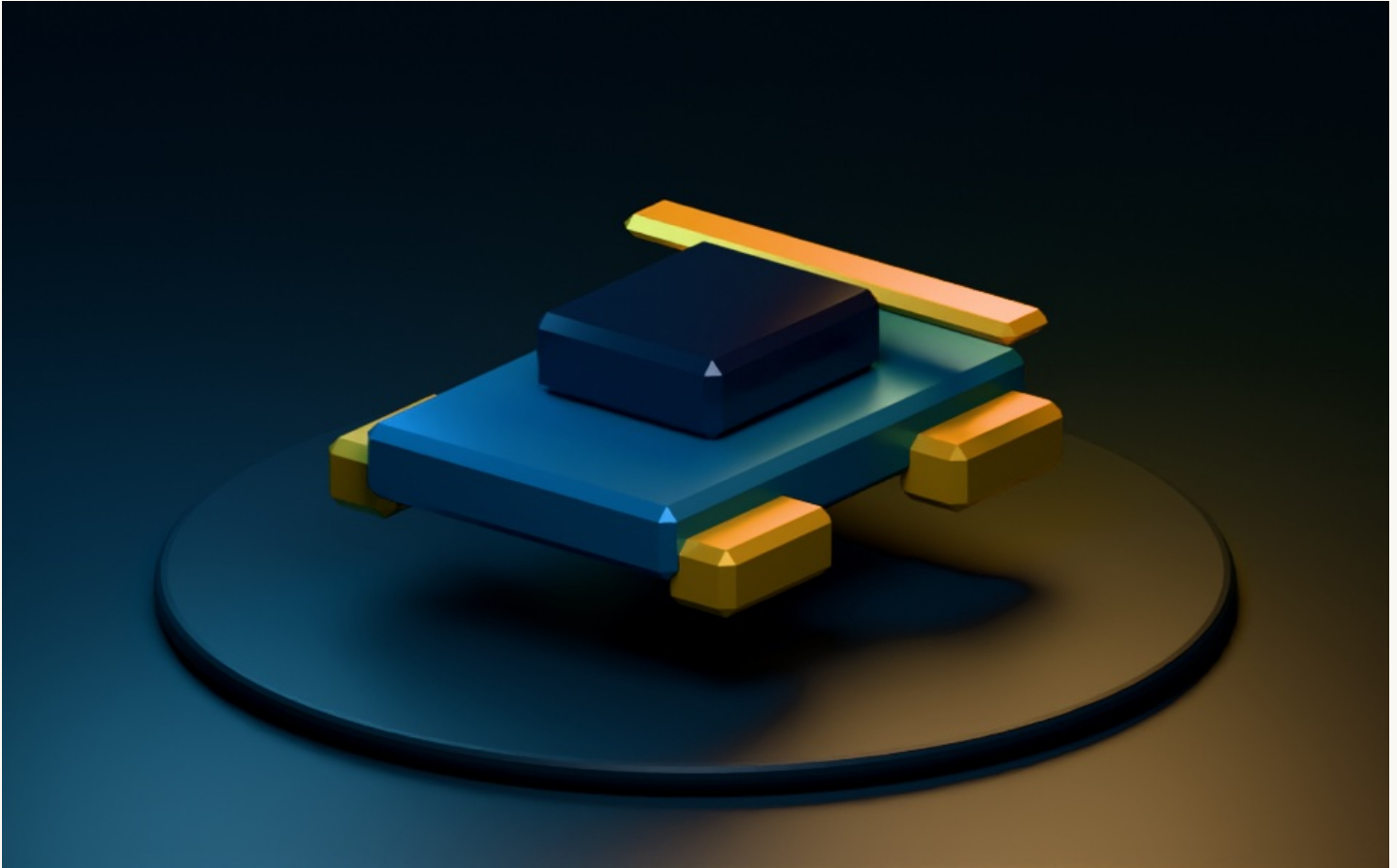


AI Builder: HOVER ONE

MONDAY / 1:35-2:35 / SESSION 1

Build your first 3D vehicle.



YOUR MISSION

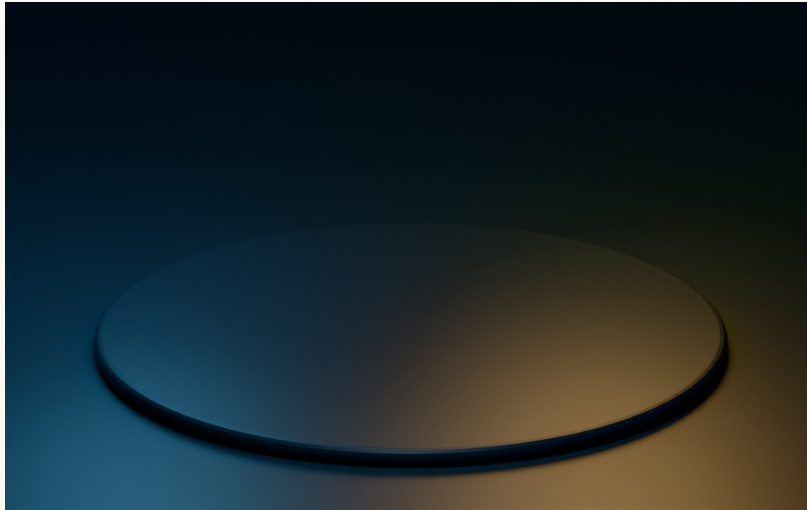
Make an original hover-car in Blender. Save a picture of it today. Next session, make this same car move.

Inspired by Rocket League vehicle showcases. This is an original model, not a game mod. The picture is the teacher-built example.

BUILD IN BLENDER / ASK VIRGIL FOR HELP / KEEP BOTH FILES

Your project: HOVER ONE

Build an original hover-car in Blender. Make a poster image of it today. Next time, we will make this same car move. Rocket League is the connection: artists create a vehicle model before it can appear in a game. This is your own design, not a Rocket League mod.



ACTUAL BLENDER RENDER / YOUR STARTING POINT

1. Open Blender 5.2. Close its welcome picture by clicking outside it. Choose File > Open. Find the parent-checked Hover-Studio.blend in Downloads, select it, and click Open Blender File. If asked, do not enable Python scripts; this scene needs none.
2. You should see an empty round stage. The camera, lights and paint colours are ready. There is no car in the starting file. If your view is not inside a camera frame, use View > Cameras > Active Camera from the menu above the large 3D area.
3. Choose File > Save As. Open Documents. Use the New Folder button to create Atticus-AI-BUILDER, then open that folder. Name the file Hover-One-v01.blend. Click Save As and confirm if asked. Press Ctrl+S after each finished part.

WHAT YOU SHOULD SEE

Your project is now in Documents > Atticus-AI-BUILDER. The starter stays unchanged.

Download the starting file in Virgil's Monday AI Builder lesson.

Learn the few controls you need

A model is a 3D object you can turn around. You will make seven parts: a body, a dark window cover, four engines and a rear wing. The picture shows the body and window cover. It is a real render of the lesson model, not a picture of Blender's buttons.

1. Keep the mouse pointer over the large 3D picture before using a shortcut. The top-left menu should say Object Mode. Click a part to select it; an orange outline means that part is selected. Ctrl+Z undoes a mistake.
2. Shift+A opens the Add menu. Choose Mesh > Cube. A new cube starts at the centre. F2 lets you name it. Press Enter after typing its name.
3. Press N over the 3D picture. In the panel that opens on the right, select Item. Under Transform, Location means where the part sits; Dimensions means its size. X goes across, Y goes front to back, and Z goes up. Double-click a number, type its replacement, then press Enter. Copy the supplied settings; no calculations are needed.
4. After setting a part's size, press Ctrl+A and choose Scale. This keeps its new shape while resetting its size controls. Press F3, type Add Bevel Modifier, and choose that result. A bevel trims sharp edges. Use its default 0.1 m amount and 1 segment. You can check them under the wrench icon on the right.

WHAT YOU SHOULD SEE

Example: Dimensions X=3, Y=4.4, Z=0.6 makes a wide, low body. Location Z=1.1 lifts it above the stage.

A small control guide

Two groups of boxes do different jobs.

LOCATION = WHERE IT SITS

X: across the stage. Y: front to back. Z: up from the stage. Change these when you want to move a part.

DIMENSIONS = HOW BIG IT IS

X: width. Y: length. Z: height. Change these when you want to reshape a part.

The body: a complete example

SETTING	X	Y	Z
Dimensions	3	4.4	0.6
Location	0	0	1.1

Double-click one field, type the number, press Enter. Repeat for the next field. Use Dimensions, not Scale, for the size numbers above.

The moves you will repeat

Add: Shift+A > Mesh > Cube

Name: F2 > type the name > Enter

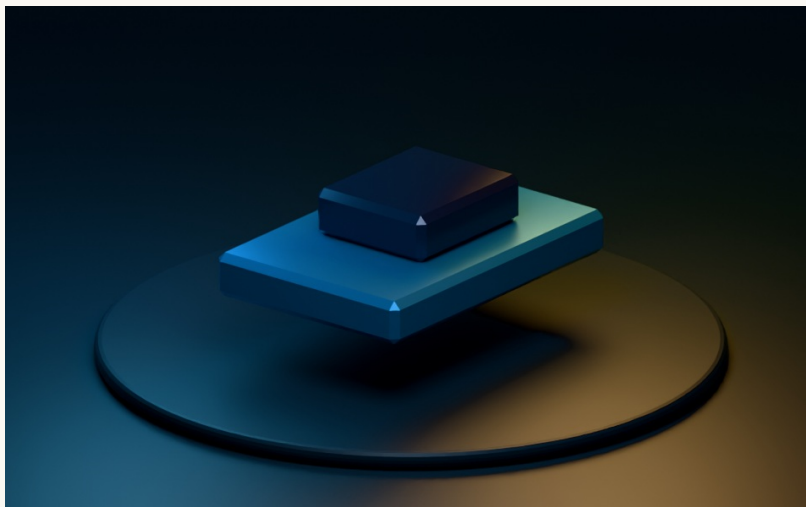
Copy: Shift+D > Esc > change Location

Trim edges: Ctrl+A > Scale; F3 > Add Bevel Modifier

Keep work: Ctrl+S

Build the body and window cover

Follow the controls you just learned. Finish one part before adding the next. The dark cover over the driver's seat is called the canopy.



ACTUAL BLENDER RENDER / THE TAUGHT MODEL

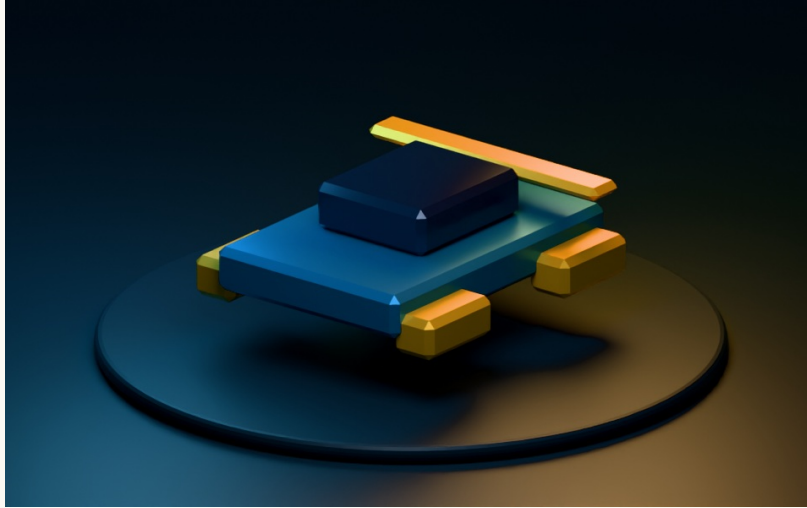
1. Add a cube and name it Body. In Item > Transform, set Dimensions to X 3, Y 4.4, Z 0.6. Set Location to X 0, Y 0, Z 1.1. Apply Scale and add the bevel, using the steps on the previous page.
2. Add a new cube and name it Canopy. Set Dimensions to X 1.8, Y 2.1, Z 0.65. Set Location to X 0, Y 0.15, Z 1.72. Apply Scale and add the bevel.
3. Look at the picture: the smaller cover sits on top of the larger body. Your shapes may still be grey. That is fine; paint comes after the engines. Press Ctrl+S.

WHAT YOU SHOULD SEE

You have two separate parts. Selecting Canopy should outline only the smaller part.

Add four engines and a wing

The engines sit beside the body. Make the first engine once, then copy it. A copy is called a duplicate. Copying saves time and keeps matching parts the same size.



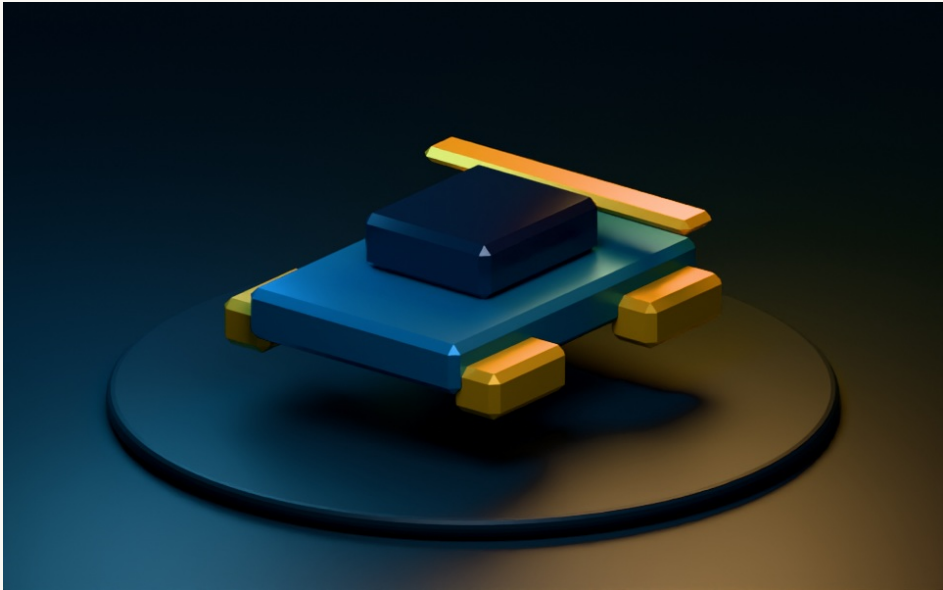
ACTUAL BLENDER RENDER / THE TAUGHT MODEL

1. Add a cube. Name it Front left. Set Dimensions to X 0.65, Y 1.35, Z 0.55. Set Location to X -1.65, Y -1.4, Z 0.85. Apply Scale and add the bevel.
2. With that engine selected, press Shift+D, then immediately press Esc. This makes a copy without moving it yet. Press F2 and name it Front right. In the Item panel, change only Location X to 1.65.
3. Copy Front right with Shift+D, then Esc. Name the copy Back right. Change only Location Y to 1.4. Copy Back right in the same way; name it Back left and change only Location X to -1.65. Do not add another bevel to copies: they already have one.
4. Add a new cube. Name it Rear wing. Set Dimensions to X 3.8, Y 0.55, Z 0.18. Set Location to X 0, Y 1.7, Z 1.8. Apply Scale and add the bevel. Press Ctrl+S.

WHAT YOU SHOULD SEE

Count seven named car parts in the list at the top right. The studio items are extra; leave those alone.

Your seven-part build map



Numbers below are X / Y / Z. Copy them into the named fields. All parts keep Rotation at 0 / 0 / 0.

PART	DIMENSIONS	LOCATION
Body	3 / 4.4 / 0.6	0 / 0 / 1.1
Canopy	1.8 / 2.1 / 0.65	0 / 0.15 / 1.72
Front left	0.65 / 1.35 / 0.55	-1.65 / -1.4 / 0.85
Front right	same as first engine	1.65 / -1.4 / 0.85
Back right	same as first engine	1.65 / 1.4 / 0.85
Back left	same as first engine	-1.65 / 1.4 / 0.85
Rear wing	3.8 / 0.55 / 0.18	0 / 1.7 / 1.8

Paint: Body = Racing blue. Canopy = Dark canopy. All engines and wing = Gold trim.

Paint it, then ask AI for useful help

A material is the paint and surface finish of a part. The starter includes ready-made blue paint, dark glass and gold trim. You will assign them to your own shapes.

1. Select Body. In the tall Properties panel on the right, click Material Properties: the red sphere icon near the bottom. Below the empty material-slot box, click the small material sphere/dropdown beside New. Choose 01 Racing blue. Do not press New.
2. Repeat for Canopy, choosing 02 Dark canopy. Select each of the four engines and Rear wing in turn, choosing 03 Gold trim for each. If the colours are hidden, press Z over the 3D area and choose Material Preview from the pie menu. Wait for it to load.
3. Return to Virgil in the browser. Say: “I am making Hover One in Blender 5.2. My goal is a clear game-poster image. My car has a blue body, a dark canopy and four gold engines. Give me one small visual improvement I can make in five minutes. Tell me the exact Blender control. Keep the same project and software.”
4. If a suggestion is unclear, say: “Explain that in ordinary words and give me only the first click.” You can describe your screen or share a screenshot through an available upload control. Virgil cannot see Blender automatically. Make one change you understand, then check the car again. If it looks worse, use Ctrl+Z.

WHAT YOU SHOULD SEE

A useful AI request says what you are making, what you see, what you want changed, and the help you need. Keep one suggestion that actually improves the picture.

Make the poster and keep your work

Render means “make the finished picture using the camera and lights.” The .blend file keeps the editable car. The .png file is the picture you can show. You need both.

1. Press Ctrl+S. Check that your parent has set Render Properties > Render Engine to Cycles and Device to CPU. From the top menu choose Render > Render Image. Wait for the picture. Use this menu instead of F12 if your laptop intercepts function keys.
2. In the finished Render Result window, choose Image > Save As. Open Documents > Atticus-AI-Builder. Set the name to Hover-One-poster.png and File Format to PNG. Click Save As Image. Saving the Blender file alone does not save this picture.
3. Close the render window using its X to return to the model. Choose File > Save As and save a second version named Hover-One-v02.blend in the same folder. Keep v01 as your earlier copy.
4. Choose File > Open, select Hover-One-v02.blend and open it. Check that the seven parts and their colours are still there. Then use Windows File Explorer to open Hover-One-poster.png. Ask a parent to copy the whole Atticus-AI-Builder folder into the family’s existing backup location.

WHAT YOU SHOULD SEE

You can reopen the editable car and its finished picture. GitHub is not needed today; no new account or public upload is part of this lesson.

Show what you made

Now the teaching and building are complete. Show your work on the screen; you do not need to write a page of answers.

1. Show the saved poster. Point to the body, canopy, four engines and wing. Explain one thing you learned to do in Blender.
2. Show the change you tried after asking Virgil. Say what looked better, or why you undid it. AI's idea is a suggestion; your picture is the test.
3. Open your saved .blend file and select one engine. This shows that you saved a model you can keep editing, not just a picture.
4. Next session: group the car's parts and animate a short hover-and-turn shot. After that: add engine light, sound and a title to make a short reveal trailer. Keep today's files. Do not start a new project.

WHAT YOU SHOULD SEE

Stop at 2:35. If setup or rendering took too long, save the model and finish the poster next session. Do not rush into animation today.

Stuck? Start here.

A shortcut does nothing

Move the pointer over the large 3D picture and try again. Use the menus if your laptop intercepts an F-key.

The wrong part changes

Press Ctrl+Z. Select the correct name in the top-right object list. Check its orange outline before changing it.

A copy seems to disappear

Shift+D then Esc puts it on top of the original. Change its Location as directed to separate them.

I cannot see the car

Choose View > Cameras > Active Camera. If the Item panel covers the view, press N over the 3D area to hide it.

I cannot find the material list

Select a car part first. Click the red Material Properties sphere, then the small dropdown next to New. Search for Racing blue or Gold trim.

The picture is grey or slow

Use Z > Solid while building. Render Image makes the final coloured picture. Save before rendering; if it fails, keep the model and ask a parent for the error message.

I do not know what to ask Virgil

Say: "I am on step __. I clicked __. I expected __, but I see __. Give me one next click in Blender 5.2."

Parent: prepare the studio

1. Install the correct Blender

Before the 60-minute block: use blender.org/download and choose Windows ARM for his Snapdragon X Elite laptop, version 5.2 LTS or newer. Avoid the Windows x64 download. A parent handles installation. Blender works locally without a child account or payment. No Fortnite, Roblox, GitHub or new AI account is required.

2. Set the tested render route

Parent render setup: in the Properties panel, click Render Properties (the camera icon). Set Render Engine to Cycles and Device to CPU. Under Sampling > Render, set Max Samples to 16 and enable Denoise. In Output Properties (printer icon), set Resolution X to 960, Y to 600, and percentage to 100. Save this prepared starter with Ctrl+S. Use these settings for today's small poster; the supplied file initially opens with Eevee. The remote Eevee render timed out; the Cycles CPU version was successfully rendered.

3. Check on his laptop

Download Hover-Studio.blend from the lesson. Open it in Blender. Confirm the stage appears and Render > Render Image produces a lit empty stage. Close the render, add a temporary cube, save a temporary test file, close and reopen it. Then reopen the untouched starter for Atticus. These checks must happen on his laptop before teaching; the remote scene test cannot prove his graphics driver works.

4. Keep setup outside class

Close extra browser tabs: his earlier screenshot showed 81% memory use. Use the 960 x 600 render settings. If Blender cannot open or render, stop setup and record the exact message. Do not make him debug an installation during his lesson. No Windows-device tests performed on Atticus's own laptop. The studio and seven-part model are tested with Blender 5.2 in a separate environment.

Parent: observe the build

Today teaches creating, sizing, copying and colouring 3D parts; using AI for a specific improvement; rendering; and keeping editable work. The starter supplies only a stage, camera, lights and materials. The car in the pictures is a teacher-built example. This is not an in-game vehicle, working mod or playable game.

Observe, do not grade speed. Suggested 10 points: creates body/canopy/engine/wing shapes 3; uses duplicate and materials 2; tries and judges one AI suggestion 2; saves and reopens editable file 2; exports poster 1. Record unfinished work separately from technical setup failures. Monday remains ungraded until work is seen; preserve Friday's 90%.

Use the existing parent-managed Virgil access for coaching. If a later project needs an outside AI service, check that service's current age rules and arrange parent-operated use before assigning it. Never assume consent alone permits an under-age account.

Sources checked October 5, 2026: [blender.org/download](https://www.blender.org/download/); code.blender.org/2025/08/blender-for-windows-on-arm; Blender manual sections on transforms, bevel modifiers, material assignment, saving files and saving rendered images. The instructions use Blender 5.2; wording may differ in later versions.

THE NEXT SESSION

Continue with Hover-One-v02.blend. Group the seven parts, animate a short hover-and-turn shot, then add lighting, sound and a title in a later session. Do not restart with a different project.

Blender download: <https://www.blender.org/download/>

Windows on Arm background: <https://code.blender.org/2025/08/blender-for-windows-on-arm/>